

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409829
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Marsolek; John et al.

Suppressing switching operation of a machine from an electric mode to an engine mode

Abstract

A control system may include a control of a user interface and a controller. The controller may be configured to cause a machine to operate in a first mode, the machine being configured for hybrid operation in the first mode using a battery without using an internal combustion engine or in a second mode using the internal combustion engine when a level of the battery is below a threshold. The controller may be configured to receive an input, via the control, indicating that the second mode of the machine is to be disabled. The controller may be configured to cause, based on the input and when the level of the battery is below the threshold, suppression of operation of the machine in the second mode.

Inventors:	Marsolek; John (Watertown, MN), Nelson; Ryan J. (Maple Grove, MN)
Applicant:	Caterpillar Paving Products Inc. (Brooklyn Park, MN)
Family ID:	91582669
Assignee:	Caterpillar Paving Products Inc. (Brooklyn Park, MN)
Appl. No.:	18/151808
Filed:	January 09, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240227774 A1	Jul. 11, 2024

Publication Classification

Int. Cl.: B60W20/13 (20160101); B60W20/15 (20160101); B60W50/14 (20200101)

U.S. Cl.:

CPC **B60W20/13** (20160101); **B60W20/15** (20160101); **B60W50/14** (20130101);
B60W2510/244 (20130101); B60W2556/45 (20200201); E01C2301/00 (20130101)

Field of Classification Search

CPC: B60W (20/13); B60W (20/15); B60W (50/14); B60W (2510/244); B60W (2556/45);
E01C (2301/00)

USPC: 701/22

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6697717	12/2003	Shioda et al.	N/A	N/A
8615342	12/2012	Suganuma et al.	N/A	N/A
9994120	12/2017	Yu	N/A	B60L 53/62
10155513	12/2017	Pietron et al.	N/A	N/A
10449953	12/2018	Choi	N/A	N/A
10611258	12/2019	Symanow et al.	N/A	N/A
10829108	12/2019	Kim et al.	N/A	N/A
10894482	12/2020	Follen et al.	N/A	N/A
11519353	12/2021	Hesketh	N/A	F02M 26/47
11669089	12/2022	Urano	701/36	B60W 10/30
2015/0217779	12/2014	Schmoll Genannt Eisenwerth et al.	N/A	N/A
2021/0094438	12/2020	Ciccone et al.	N/A	N/A
2021/0139014	12/2020	Zhao	N/A	B60W 30/188
2021/0156351	12/2020	Harmeyer et al.	N/A	N/A
2021/0162883	12/2020	Shinohara et al.	N/A	N/A
2022/0089142	12/2021	Yokoyama et al.	N/A	N/A
2023/0087055	12/2022	Berkooz	701/22	B60W 50/14
2023/0264678	12/2022	Ogawa	701/22	H01M 10/425

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102011002306	12/2011	DE	N/A
102016206743	12/2016	DE	N/A
3333036	12/2020	EP	N/A
3839436	12/2020	EP	N/A
2000204996	12/1999	JP	N/A
3092403	12/1999	JP	N/A

Primary Examiner: Rudy; Andrew Joseph

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to hybrid electric machines and, for example, to suppressing switching operation of a machine from an electric mode to an engine mode.

BACKGROUND

(2) A machine may include a battery that powers systems, such as a propulsion system and a work implement system, of the machine to enable operation of the machine in a fully electric mode (e.g., without use of an internal combustion engine). The machine may also include an internal combustion engine. When a charge level of the battery is below a threshold, the engine may automatically turn on to charge the battery or power the machine directly, thereby facilitating uninterrupted operation of the machine. Such a machine capable of operating in a fully electric mode and using an internal combustion engine may be referred to as a “hybrid electric machine.” Automatic starting of an internal combustion engine of a machine may prevent operator control over emissions produced by the machine, may consume excessive fuel, may increase wear to the engine, or the like. Moreover, in some examples, a machine that is used at a particular work site may be required to produce zero emissions, and automatic starting of an internal combustion engine produces emissions in contravention of that requirement even though the machine may be capable of zero-emission operation.

(3) U.S. Pat. No. 10,829,108 (the '108 patent) discloses controlling engine operation of a hybrid electric vehicle. In particular, the '108 patent discloses that when it is determined that an engine or a catalytic converter is cold, that a vehicle is parked in an indoor garage, and that a state of charge of a battery is enough for an electric vehicle mode, a controller does not start the engine in response to a driver's command to start the vehicle and the vehicle drives without operating the engine. However, the '108 patent does not disclose suppressing operation of the engine when the battery level is below a threshold (e.g., not enough for the electric vehicle mode).

(4) The control system of the present disclosure solves one or more of the problems set forth above and/or other problems in the art.

SUMMARY

(5) A control system may include a control of a user interface and a controller. The controller may be configured to cause a machine to operate in a first mode, the machine being configured for hybrid operation in the first mode using a battery without using an internal combustion engine or in a second mode using the internal combustion engine when a level of the battery is below a threshold. The controller may be configured to receive an input, via the control, indicating that the second mode of the machine is to be disabled. The controller may be configured to cause, based on the input and when the level of the battery is below the threshold, suppression of operation of the machine in the second mode.

(6) A method may include causing, by a controller, a machine to operate in a first mode, the machine being configured for hybrid operation in the first mode using a battery without using an internal combustion engine or in a second mode using the internal combustion engine when a level of the battery is below a threshold (as used herein, the term “level” refers to a charge level of a battery, as determined in any appropriate manner). The method may include determining, by the controller, that the level of the battery is below the threshold. The method may include selectively causing, by the controller, the machine to switch from operating in the first mode to operating in the second mode based on whether an input indicating that the second mode is to be disabled is received.

(7) A machine may include a battery and an internal combustion engine. The machine may be configured for hybrid operation in a first mode using the battery without using the internal combustion engine or in a second mode using the internal combustion engine when a level of the battery is below a threshold. The machine may include a controller configured to cause the machine to operate in the first mode. The controller may be configured to receive an input, via a user

interface associated with the machine, indicating that the second mode of the machine is to be disabled. The controller may be configured to cause, based on the input and when the level of the battery is below the threshold, suppression of operation of the machine in the second mode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating an example machine.
- (2) FIG. 2 is a diagram illustrating an example control system.
- (3) FIG. 3 is a flowchart of an example process associated with suppressing switching operation of a machine from an electric mode to an engine mode.

DETAILED DESCRIPTION

- (4) This disclosure relates to a control system, which is applicable to any machine capable of hybrid operation in a first mode using a battery without using an internal combustion engine or in a second mode using the internal combustion engine. For example, the machine may be a paving machine, a compactor machine, a cold planer, a grading machine, a backhoe loader, a wheel loader, a harvester, an excavator, a motor grader, a skid steer loader, a tractor, a dozer, a vehicle, or the like.
- (5) FIG. 1 is a diagram illustrating an example machine **100**. In FIG. 1, the machine **100** is depicted as a paving machine, however, the machine **100** may be another type of machine, as described herein.
- (6) The machine **100** includes a frame **102**, and a set of ground-engaging elements **104**, such as tracks and/or wheels, coupled with the frame **102**. The machine **100** may be configured for hybrid operation. For example, the machine **100** may be a hybrid electric machine (e.g., that includes a hybrid drivetrain) that combines an internal combustion engine **106** (e.g., a diesel engine or a gasoline engine) with an electric propulsion system (e.g., including one or more electric motors) powered by a battery **108**. The engine **106** and the battery **108** are supported by the frame **102**. The ground-engaging elements **104** may be driven by the engine **106** and/or the electric propulsion system (e.g., concurrently or at different times). The engine **106** and/or the electric propulsion system may power other systems of the machine **100**, such as a work implement. In some implementations, the machine **100** includes an electrical generator **110** (i.e., an alternator) coupled to the engine **106** and the battery **108**. The engine **106** may drive the electrical generator **110** to charge the battery **108**.
- (7) The machine **100** includes a screed assembly **112**. The screed assembly **112** may be positioned at a rear end of the machine **100** to spread and compact paving material (e.g., asphalt) into a mat **114**. The machine **100** includes an operator station **116** having a seat and a console, which may include various controls for directing operations of the machine **100**. For example, the operator station **116** may include a touchscreen capable of displaying a user interface that includes one or more virtual controls (e.g., buttons, switches, levers, or the like). Additionally, or alternatively, a user interface of the operator station **116** may include one or more mechanical controls.
- (8) The machine **100** includes a controller **118** (e.g., an electronic control module (ECM)). The controller **118** may be a hybrid operation controller of the machine **100**. The controller **118** may include one or more memories and one or more processors that implement operations associated with hybrid operation of the machine **100**, as described in connection with FIG. 2.
- (9) As indicated above, FIG. 1 is provided as an example. Other examples may differ from what is described with regard to FIG. 1.
- (10) FIG. 2 is a diagram illustrating an example control system **200**. The control system **200** may include the controller **118** and a control **202**. The control **202** may be included in a user interface (e.g., a mechanical user interface or a digital user interface). The control system **200** may provide control of hybrid operation of the machine **100**. In hybrid operation, the machine **100** may operate

in a first mode (referred to herein as an “electric mode”) using the battery **108** without using the engine **106** (e.g., the battery **108** may be used to power an electric propulsion system) or in a second mode (referred to herein as an “engine mode”) using the engine **106** (e.g., the engine **106** is turned on to charge the battery **108** and/or power the machine **100** directly) when a level (e.g., a charge level) of the battery **108** is below a threshold. In other words, if the battery **108** is sufficiently charged, then the machine **100** may operate in the first mode. Otherwise, the machine **100** may operate in the second mode. The controller **118** (e.g., the one or more memories and/or the one or more processors of the controller **118**) may be configured to perform operations described in FIG. 2.

(11) When the machine **100** is started, the controller **118** may cause the machine **100** to operate in the electric mode (provided that the level of the battery **108** is equal to or above the threshold). In some implementations, the controller **118** may monitor an amount of time (e.g., a quantity of machine hours) that the machine **100** operates in the electric mode. For example, the controller **118** may monitor the amount of time that the machine **100** operates in the electric mode and a total amount of time (e.g., a quantity of machine hours) that the machine **100** is operating.

(12) The controller **118** may receive an input indicating that the engine mode of the machine **100** is to be disabled. The controller **118** may receive the input via a user interface associated with the machine **100** (e.g., the input may be a manual input). For example, the controller **118** may receive the input via the control **202**. In some examples, the user interface (e.g., the control **202**) may be on board the machine **100** (e.g., in operator station **116**). In some other examples, the user interface (e.g., the control **202**) may be on a device remote from the machine **100** (e.g., a back office device, a user device associated with a supervisor, or the like).

(13) The input indicating that the engine mode is to be disabled may be provided by an individual associated with the machine **100** (e.g., an operator, a supervisor, or the like) via the control **202** (e.g., a button, a switch, a lever, or the like). The individual may provide the input indicating that the engine mode is to be disabled because the machine **100** is operating in an area (e.g., a work site) associated with a requirement of zero emissions. Thus, when the engine mode is disabled, the machine **100** may be considered an electric-only machine or a zero emissions machine, such that the machine **100** may operate in the area associated with the requirement of zero emissions.

(14) Responsive to receiving the input indicating that the engine mode is to be disabled, the controller **118** may perform one or more actions to record that the engine mode is to be disabled. For example, the controller **118** may set a flag indicating that the engine mode is disabled. In some implementations, the controller **118** may receive, via the user interface, an indication of a target proportion of time (e.g., a target proportion of machine hours) that the machine **100** is to operate in the electric mode. The controller **118** may receive the indication of the target proportion of time contemporaneously with, prior to, or after the input indicating that the engine mode is to be disabled.

(15) The controller **118** may determine that the level of the battery **108** is below the threshold for switching from operating in the electric mode to operating in the engine mode. For example, the threshold may represent a charge level of the battery **108** at which the battery **108** has insufficient charge to support operation of the machine **100** in the electric mode. The controller **118** may be configured to provide a command to cause the machine **100** to switch from operating in the electric mode to operating in the engine mode when the level of the battery **108** is below the threshold.

(16) The controller **118** may selectively cause the machine to switch from operating in the electric mode to operating in the engine mode based on whether the input indicating that the engine mode is to be disabled is received. If the input indicating that the engine mode is to be disabled is not received by the controller **118**, then the controller **118** may cause switching from the electric mode to the engine mode when the level of the battery **108** is below the threshold. If the input indicating that the engine mode is to be disabled is received by the controller **118**, then the controller **118** may not cause switching (e.g., may refrain from causing switching) from the electric mode to the engine

mode when the level of the battery **108** is below the threshold. For example, the controller **118** may cause, based on the input indicating that the engine mode is to be disabled and when the level of the battery **108** is below the threshold, suppression of operation of the machine **100** in the engine mode. To suppress operation of the machine **100** in the engine mode, the controller **118** may block, ignore, or refrain from providing a command to switch from the electric mode to the engine mode, which the controller **118** may be configured to otherwise provide when the level of the battery **108** is below the threshold. In this way, emissions of the machine **100** may be suppressed.

(17) In some implementations, responsive to receiving the input indicating that the engine mode is to be disabled, the controller **118** may determine an estimated time (e.g., a time duration or a time of day and date) at which the level of the battery **108** is to be below the threshold. The controller **118** may transmit, prior to the estimated time, a request for recharging the battery **108**. As described below, such recharging can be performed by components other than the engine **106**, meaning the engine mode can remain disabled. Additionally, or alternatively, the controller **118** may transmit the request for recharging the battery **108** responsive to detecting that the level of the battery is below a recharging threshold (e.g., that is higher than the threshold for switching from the electric mode to the engine mode). Additionally, or alternatively, based on causing suppression of the engine mode, the controller **118** may transmit the request for recharging of the battery **108**. Additionally, or alternatively, the controller **118** may transmit the request for recharging the battery **108** responsive to receiving an indication (e.g., a manual indication provided via the user interface) to transmit the request for recharging the battery **108**. The request may be for immediate recharging of the battery **108** or for recharging of the battery **108** at a future time that is prior to the estimated time.

Moreover, the request may indicate an identifier of the machine **100**, a location of the machine **100**, and/or the estimated time, among other examples. The controller **118** may transmit the request to a device remote from the machine **100** (e.g., a back office device). The request may cause a charging rover (e.g., an autonomous charging rover) or a non-autonomous charging vehicle to perform recharging of the battery **108**.

(18) In some implementations, the controller **118** may receive, from a device remote from the machine **100** (e.g., a back office device), an indication to operate the machine **100** in the engine mode after suppression of the engine mode. In other words, the indication may override a previous suppression of the engine mode. The controller **118** may cause, based on the indication, operation of the machine **100** in the engine mode. In this way, the engine **106** may be started in case of an emergency and/or if the machine **100** is unable to be moved solely by electrical power (e.g., if the machine **100** is down due to high voltage electrical system failure).

(19) The control **202** used to provide the input indicating that the engine mode is to be disabled (or a different control) may also enable an input indicating that the engine mode is to be enabled. For example, an individual may toggle between disabling the engine mode and enabling the engine mode using the control **202**. In some implementations, enabling of the engine mode may be permitted at any time after the engine mode is disabled. Alternatively, enabling of the engine mode may be permitted after the engine mode is disabled provided that a condition (e.g., one or more conditions) for permitting the engine mode to be enabled is satisfied. Here, until the condition is satisfied, the controller **118** may disable the control **202** used to provide an input indicating that the engine mode is to be enabled. Additionally, or alternatively, until the condition is satisfied, the controller **118** may reject an input indicating that the engine mode is to be enabled.

(20) The controller **118** may determine that the condition for permitting the engine mode to be enabled is satisfied. The condition may be that a particular time period has elapsed since suppression of operation of the machine **100** in the engine mode, that the machine **100** has traveled at least a particular distance since suppression of operation of the machine **100** in the engine mode, that a location of the machine is outside of an area associated with a zero emissions requirement (e.g., a work site associated with electric-only operation), and/or that a credential (e.g., a password, a digital certificate, or the like), indicating authorization to enable operation of the machine **100** in

the second mode, is received, among other examples. Based on satisfaction of the condition, the controller **118** may enable a subsequent input (e.g., by enabling the control **202** used for providing the subsequent input) and/or accept a subsequent input (e.g., a subsequent input provided via the control **202**) indicating that the engine mode of the machine **100** is to be enabled.

(21) As described herein, the controller **118** may monitor (e.g., collect and store data) an amount of time that the machine **100** operates in the electric mode, which may indicate whether the machine **100** (or a fleet of machines that includes the machine **100**) is meeting an emissions target. The controller **118** may determine that the amount of time is indicative of non-compliance with a standard (e.g., a zero emissions standard, a low emissions standard, an electric mode usage standard, or the like). For example, if the machine **100** operates in the electric mode for less than a threshold amount of an uptime of the machine **100**, then the controller **118** may determine that the amount of time is indicative of non-compliance with the standard. The controller **118** may transmit (e.g., to a device remote from the machine **100**, such as a back office device) a notification indicating non-compliance with the standard, thereby enabling remedial actions to enforce compliance with the standard.

(22) The controller **118** may provide, via the user interface and based on the amount of time, information indicating whether a proportion of time that the machine **100** has operated in the electric mode meets the target proportion of time that was indicated (e.g., indicating a status of the machine **100** as to meeting the target proportion of time). For example, the information may indicate a total time in which the machine **100** has operated in the electric mode, the proportion of time that the machine **100** has operated in the electric mode, and/or the target proportion of time. Additionally, the information may indicate an amount of fuel savings and/or an amount of emissions savings attributable to operation of the machine **100** in the electric mode. Moreover, the information may indicate a comparison of the amount of time that the machine **100** has operated in the electric mode to amounts of time associated with one or more other machines in a fleet with the machine **100**. The controller **118** may also provide the information to a device remote from the machine **100** (e.g., a back office device or a device implementing a database).

(23) The controller **118** may determine that the proportion of time that the machine **100** has operated in the electric mode indicates that the machine **100** will be unable to meet (e.g., is not on track to meet) the target proportion of time. In some implementations, based on determining that the machine **100** will be unable to meet the target proportion of time, the controller **118** may automatically disable the engine mode and/or may prevent enabling of the engine mode after the engine mode is disabled. Additionally, or alternatively, based on determining that the machine **100** will be unable to meet the target proportion of time, the controller **118** may select the machine **100**, from a fleet of machines, for operating in an area (e.g., a work site) associated with a requirement of zero emissions, to thereby increase a time of electric mode operation of the machine **100**.

(24) In this way, the control system **200** improves operator control over starting of the engine **106**, thereby improving operator control over emissions produced by the machine **100**.

(25) As indicated above, FIG. 2 is provided as an example. Other examples may differ from what is described with regard to FIG. 2.

(26) FIG. 3 is a flowchart of an example process **300** associated with suppressing switching operation of a machine from an electric mode to an engine mode. One or more process blocks of FIG. 3 maybe performed by a controller (e.g., controller **118**). Additionally, or alternatively, one or more process blocks of FIG. 3 maybe performed by another device or a group of devices separate from or including the controller, such as another device or component that is internal or external to the machine **100**.

(27) As shown in FIG. 3, process **300** may include causing a machine to operate in a first mode, the machine being configured for hybrid operation in the first mode using a battery without using an internal combustion engine or in a second mode using the internal combustion engine when a level of the battery is below a threshold (block **310**). For example, the controller may cause a machine to

operate in a first mode, as described above. Process **300** may include receiving an input indicating that the second mode is to be disabled (block **312**).

(28) Process **300** may include determining, responsive to receiving the input, an estimated time at which the level of the battery is to be below the threshold, and transmitting, prior to the estimated time, a request for recharging of the battery. Process **300** may include transmitting, responsive to detecting that the level of the battery is below a recharging threshold, a request for recharging of the battery.

(29) As further shown in FIG. 3, process **300** may include determining that the level of the battery is below the threshold (block **320**). For example, the controller may determine that the level of the battery is below the threshold, as described above.

(30) As further shown in FIG. 3, process **300** may include selectively causing the machine to switch from operating in the first mode to operating in the second mode based on whether an input indicating that the second mode is to be disabled is received (block **330**). For example, the controller may selectively cause the machine to switch from operating in the first mode to operating in the second mode based on whether an input indicating that the second mode is to be disabled is received, as described above. Selectively causing the machine to switch from operating in the first mode to operating in the second mode may include causing, based on receiving the input indicating that the second mode is to be disabled and when the level of the battery is below the threshold, suppression of operation of the machine in the second mode (block **332**).

(31) Process **300** may include receiving, from a device remote from the machine, an indication to operate the machine in the second mode after suppression of operation of the machine in the second mode, and causing, based on the indication, operation of the machine in the second mode (block **334**). Process **300** may include determining that a condition for permitting the second mode of the machine to be enabled is satisfied, and enabling or accepting, based on satisfaction of the condition, a subsequent input indicating that the second mode of the machine is to be enabled (block **336**).

(32) Process **300** may include monitoring an amount of time that the machine operates in the first mode. Process **300** may include receiving an indication of a target proportion of time that the machine is to operate in the first mode, and providing, via a user interface and based on the amount of time, information indicating whether a proportion of time that the machine has operated in the first mode meets the target proportion of time. Process **300** may include monitoring an amount of time that the machine operates in the first mode, determining that the amount of time is indicative of non-compliance with a standard, and transmitting a notification indicating non-compliance with the standard.

(33) Although FIG. 3 shows example blocks of process **300**, in some implementations, process **300** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 3. Additionally, or alternatively, two or more of the blocks of process **300** may be performed in parallel.

INDUSTRIAL APPLICABILITY

(34) The control system described herein may be used with any machine capable of hybrid operation in an electric mode using a battery without using an internal combustion engine or in an engine mode using the internal combustion engine when a level of the battery is below a threshold. For example, the control system may be used in a hybrid electric machine that is to be used in an area (e.g., a work site) associated with a requirement of zero emissions. The control system is useful for facilitating improved control over starting of the engine, thereby improving control over emissions produced by the machine. For example, the control system may cause suppression of operation of the machine in the engine mode even if the level of the battery is below the threshold, thereby reducing emissions produced by the machine and enabling use of the machine in an area associated with a requirement of zero emissions.

(35) The foregoing disclosure provides illustration and description, but is not intended to be exhaustive or to limit the implementations to the precise forms disclosed. Modifications and

variations may be made in light of the above disclosure or may be acquired from practice of the implementations. Furthermore, any of the implementations described herein may be combined unless the foregoing disclosure expressly provides a reason that one or more implementations cannot be combined. Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various implementations. Although each dependent claim listed below may directly depend on only one claim, the disclosure of various implementations includes each dependent claim in combination with every other claim in the claim set.

(36) As used herein, “a,” “an,” and a “set” are intended to include one or more items, and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

Claims

1. A control system, comprising: a control of a user interface; and a controller configured to: cause a machine to operate in a first mode using a battery without using an internal combustion engine; receive an input, via the control, indicating that a second mode of the machine is to be disabled, wherein, in the second mode, the machine uses the internal combustion engine when a level of the battery is below a threshold; and cause, based on the input and when the level of the battery is below the threshold, suppression of operation of the machine in the second mode.
2. The control system of claim 1, wherein the user interface is on a device remote from the machine.
3. The control system of claim 1, wherein the user interface is on board the machine.
4. The control system of claim 1, wherein the controller is further configured to: receive, from a device remote from the machine, an indication to operate the machine in the second mode after suppression of operation of the machine in the second mode; and cause, based on the indication, operation of the machine in the second mode.
5. The control system of claim 1, wherein the controller is further configured to: determine, responsive to receiving the input, an estimated time at which the level of the battery is to be below the threshold; and transmit, prior to the estimated time, a request for recharging of the battery.
6. The control system of claim 1, wherein the controller is further configured to: determine that the level of the battery is below the threshold prior to causing suppression of operation of the machine in the second mode.
7. The control system of claim 1, wherein the controller is further configured to: receive an indication of a target proportion of time that the machine is to operate in the first mode; monitor an amount of time that the machine operates in the first mode; and provide, via the user interface and based on the amount of time, information indicating whether a proportion of time that the machine has operated in the first mode meets the target proportion of time.
8. A method, comprising: causing, by a controller, a machine to operate in a first mode using a battery without using an internal combustion engine; determining, by the controller, that a level of the battery is below a threshold; and selectively causing, by the controller, the machine to switch from operating in the first mode to operating in a second mode based on whether an input indicating that the second mode is to be disabled is received, wherein, in the second mode, the machine uses the internal combustion engine.
9. The method of claim 8, further comprising: receiving the input indicating that the second mode is to be disabled.

10. The method of claim 9, wherein selectively causing the machine to switch from operating in the first mode to operating in the second mode comprises: causing, based on receiving the input indicating that the second mode is to be disabled and when the level of the battery is below the threshold, suppression of operation of the machine in the second mode.
 11. The method of claim 10, further comprising: receiving, from a device remote from the machine, an indication to operate the machine in the second mode after suppression of operation of the machine in the second mode; and causing, based on the indication, operation of the machine in the second mode.
 12. The method of claim 10, further comprising: determining that a condition for permitting the second mode of the machine to be enabled is satisfied; and enabling or accepting, based on satisfaction of the condition, a subsequent input indicating that the second mode of the machine is to be enabled.
 13. The method of claim 12, wherein the condition is that one or more of: a particular time period has elapsed since suppression of operation of the machine in the second mode, the machine has traveled at least a particular distance since suppression of operation of the machine in the second mode, a location of the machine is outside of an area associated with a zero emissions requirement, or a credential, indicating authorization to enable operation of the machine in the second mode, is received.
 14. The method of claim 8, further comprising: monitoring an amount of time that the machine operates in the first mode.
 15. The method of claim 14, further comprising: receiving an indication of a target proportion of time that the machine is to operate in the first mode; and providing, via a user interface and based on the amount of time, information indicating whether a proportion of time that the machine has operated in the first mode meets the target proportion of time.
 16. A machine, comprising: a battery; an internal combustion engine; and a controller configured to: cause the machine to operate in a first mode using the battery without using the internal combustion engine; receive an input, via a user interface associated with the machine, indicating that a second mode of the machine is to be disabled, wherein, in the second mode, the machine uses the internal combustion engine when a level of the battery is below a threshold; and cause, based on the input and when the level of the battery is below the threshold, suppression of operation of the machine in the second mode.
 17. The machine of claim 16, wherein the user interface is on a device remote from the machine.
 18. The machine of claim 16, wherein the controller is further configured to: receive, from a device remote from the machine, an indication to operate the machine in the second mode after suppression of operation of the machine in the second mode; and cause, based on the indication, operation of the machine in the second mode.
 19. The machine of claim 16 wherein the controller is further configured to: transmit, responsive to detecting that the level of the battery is below a recharging threshold, a request for recharging of the battery.
 20. The machine of claim 16, wherein the controller is further configured to: monitor an amount of time that the machine operates in the first mode; determine that the amount of time is indicative of non-compliance with a standard; and transmit a notification indicating non-compliance with the standard.
-